Comparing B-spline and uniGradICON-AI Foundational Deformable Image Registration Methods: Evaluating the Tradeoff Between Accuracy and Speed

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ABSTRACT

In lung microwave ablation (LMA) therapy, tumor cells are eradicated by heating tissue to cytotoxic temperatures, producing a region of cell death termed the ablation zone. The ablation margin, defined as the zone of destroyed tissue extending beyond the tumor boundary, is a primary determinant of local recurrence risk, as an inadequate margin indicates potential residual viable tumor cells. Accurate deformable registration of pre-treatment CT scans depicting the tumor with post-treatment CT scans depicting the ablation zone enables precise quantification of the ablation margin, thereby facilitating confirmation of adequate treatment coverage. In this study, we compared a B-spline–based parametric approach with uniGradICON, a foundational AI model, for deformable registration of pre- and post-treatment lung microwave ablation (LMA) CT scans. This retrospective study consisted of 114 ablations between 01/01/2015-01/01/2019, performed using a Neuwave PR probe (Ethicon, NJ). The multi-step B-spline approach employed a free-form deformation model with normalized correlation coefficient (NCC) as the similarity metric. The uniGradICON foundational AI model was evaluated with up to 200 instance optimization (IO) iterations following zero-shot inference with local normalized cross-correlation (LNCC) as the similarity metric. UniGradICON was not specifically trained for the LMA task. Tumor and ablation zones were excluded from the registrations using cost function masking to prevent them from matching to each other. Target registration error (TRE) was used to quantify the registration accuracy, and the results were compared using paired t-test. Computation efficiency was measured by the average time required per registration. The B-spline registration achieved a significantly lower TRE (0.43 mm, SD=0.34, 95% CI [0.36-0.49]) than the best-performing uniGradICON model (1.11 mm, SD=0.76, 95% CI [0.96-1.25]) with 75 IO iterations ($p<0.001$). In contrast, B-spline registration was much slower requiring >10 minutes per registration, whereas uniGradICON was 4-8 times faster averaging at 1.3 minutes with 75 IO iterations. This preliminary study highlights a potential accuracy–speed trade-off between parametric B-spline and uniGradICON foundational model, motivating a larger-scale study with more data, careful control of variables, and task specific fine tuning for a rigorous comparison.

Keywords: Deformable image registration, B-spline registration, foundational AI models, uniGradICON, lung microwave ablation, target registration error, medical image analysis

1. INTRODUCTION

In lung microwave ablation (LMA) therapy, tumor cells are eradicated by heating tissue to cytotoxic temperature, producing a region of cell death termed as the ablation zone.[1] The tumor is first localized in a pre-treatment scan acquired prior to the procedure, and thermal ablation is performed by setting the microwave power and duration based on a vendor specification to maximize the likelihood of complete tumor eradication. The post treatment follow-up scan is acquired one month later to allow optimal visualization and assessment of the ablation zone (Figure 1a).

The treatment is considered successful when the ablation zone adequately encompasses the tumor with a sufficient margin of destroyed healthy tissue defined as the ablation margin. Achieving adequate margin is necessary to prevent residual viable tumor cells, which is a primary determinant of treatment failure and post-treatment local recurrence risk.[2] A three-dimensional (3D) margin assessment allows to fully evaluate the success of the procedure, as it captures the spatial extent of tissue destruction around the tumor in all directions. Accurate deformable registration of pre-treatment CT scans

depicting the tumor with post-treatment CT scans depicting the ablation zone enables precise quantification of the ablation margin (Figure 1b), thereby facilitating confirmation of adequate treatment coverage.[3]

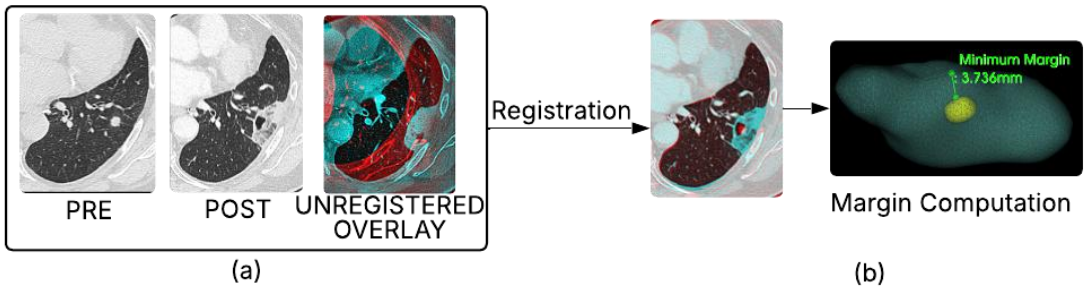


Figure 1: (a) Pre and post ablation procedure CT scans with tumor and ablation zone respectively. Also shows unregistered overlay with pre (red) and post (blue) scans. (b) Deformable registration overlay of pre and post scans; white regions indicate accurate anatomical alignment, as seen at neighboring blood vessel locations. Registration enables 3D margin assessment.

B-spline–based deformable image registration is a parametric free-form deformation method[4] in which non-rigid transformations are modeled using a sparse grid of control points interpolated by smooth cubic B-spline basis functions. These are piecewise-defined polynomial function with local support, where each voxel's displacement is governed by nearby control points enabling localized deformations while maintaining overall smoothness. The transformation parameters are estimated via iterative optimization of an image similarity metric, combined with regularization terms that penalize non-smooth or physically implausible deformations. This provides a strong balance between flexibility and stability, enabling accurate alignment of complex anatomical structures, though incurring computational cost for optimization.

More recently, AI based deformable image registration methods have gained popularity primarily due to their speed of execution. After the initial computationally intensive training phase, these models can perform registration via direct model inference in contrast to the iterative optimization of conventional pair-wise optimization methods.[5] In particular, uniGradICON is a foundational AI model based on the U-Net encoder-decoder architecture for non-parametric registrations,[6] which learns a voxel-wise displacement field by optimizing a similarity objective combined with regularization to produce smooth, anatomy-preserving deformations. The model is trained on a diverse data employing gradient-based inverse consistency to improve robustness and symmetry and supports instance-specific optimization at test time for further refinement of predicted deformation field. Since uniGradICON is not trained to handle drastic local anatomical changes as in LMA cases, targeted loss function masking is needed to ensure accurate registrations. Comparing the B-Spline and foundational AI uniGradICON provides a crucial benchmark for evaluating the trade-offs and informing their suitability for applications.

2. METHODS

This IRB-approved retrospective study consisted of 114 ablations between 01/01/2015 – 01/01/2019, performed using Neuwave PR probe (Ethicon, NJ). The data for each ablation included the pre- and post-treatment CT scans with median scanning resolution of 0.69 mm along x and y and slice thickness of 1.25 mm. This data has been utilized in prior studies examining the association of clinical variables with local recurrence[7] and preoperative prediction of lung ablation zone.[8] To prevent the tumor in pre from matching the ablation zone in the post scan, the tumor, ablation and abnormal regions were manually segmented and masked out during registration (Figure 2) via cost function masking.[9] The lung segmentation was used to crop pre- and post-treatment CT scans to the relevant lung and focus the registration on the region of interest (ROI). All manual segmentations were performed using 3D slicer software (v.5.5.0).[10]

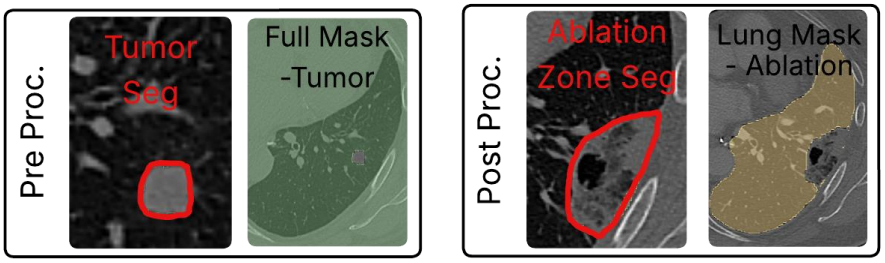


Figure 2: Ground truth segmentations of tumor and registration mask cropped to lung on pre ablation scan (left) and segmentation of ablation zone and lung mask on post ablation scan.

Our multi-step B-spline approach employed a free-form deformation model with normalized correlation coefficient (NCC) as the similarity metric.[8] The uniGradICON foundational AI[6] model utilized a composite loss function consisting of a similarity term and a regularization term. It used local normalized cross-correlation (LNCC) as the similarity term and gradient inverse consistency as the regularization term,[11] which enforced approximate invertibility of the deformation field by penalizing deviation from identity when composing forward and backward transformation. It additionally incorporated an intensity conservation loss that ensured consistent intensity adjustment during registration tasks that required mass conservation.[6] This loss penalized non-physical changes in image intensity that can result from the deformation.

The uniGradICON model was run for different number of instance optimization (IO) steps beyond zero-shot prediction, which helps refine the registration for each image pair at the test time by updating the model parameters locally. The uniGradICON model was first evaluated using the default hyperparameters (learning rate of 0.00002 and sigma of 5 for the LNCC) at IO iterations of 1, 50 and 200 post zero shot inference. Further, a grid search was performed across a range of hyperparameter values including IO steps of 20, 50, 75, 100, 125, and 150, learning rates (LR) of 0.00002, 0.00004, 0.00006, 0.00008, 0.0001, 0.0005, 0.001, and 0.005, and LNCC Gaussian kernel sigma values of 1, 2, 3, 4, and 5 to assess the best performing model. Note that hyperparameter tuning was performed only for instance optimization (IO) and did not involve retraining the base uniGradICON model. Additionally, we computed the time taken for different IO steps with the default model and the best hyper parameter tuned model.

Target registration error (TRE) was used to quantify registration accuracy. The landmark points for TRE included vessel bifurcations and were picked by a board-certified radiologist on both pre- and post-operative scans. Paired t-test was used for comparing the results between B-spline and uniGradICON registrations. Computation efficiency was measured by the average time required per registration.

All experiments were conducted on a server with an AMD EPYC 7543P processor with 32 CPU cores, 500 GB of RAM, and an NVIDIA A40 GPU with 48 GB of VRAM. The B-spline registration was run on the CPU using the ITKElastix python library.[12] The uniGradICON model was run on the GPU using the uniGradICON GitHub python library.[6] Google's Gemini[13] was used for assistance in editing the paper.

3. RESULTS

3.1 Registration Accuracy

Figure 3 shows the TRE results obtained by the different registrations. The B-spline registration method achieved the lowest TRE with a mean of 0.43 mm (SD:0.34, 95% CI [0.36-0.49] mm). In comparison, uniGradICON with default hyperparameters and 200 IO steps achieved a mean TRE of 1.2 mm (SD=0.83 mm, 95% CI [1.05-1.36 mm]). The uniGradICON model showed significant improvement in TRE with an increase in IO steps. Zero-shot prediction (IO=1) resulted in a mean TRE of 4.66 mm (SD=3.40, 95% CI [4.03-5.30 mm]), which decreased to 2.09 mm (SD=1.15, 95% CI [1.87-2.30 mm]) with 50 IO and further decreased to 1.2 mm (SD=0.83, 95% CI [1.05-1.36 mm]) with 200 IO. All pairwise comparisons between different IO iterations were statistically significant ($p<0.001$), demonstrating instance optimization helps improve registration accuracy.

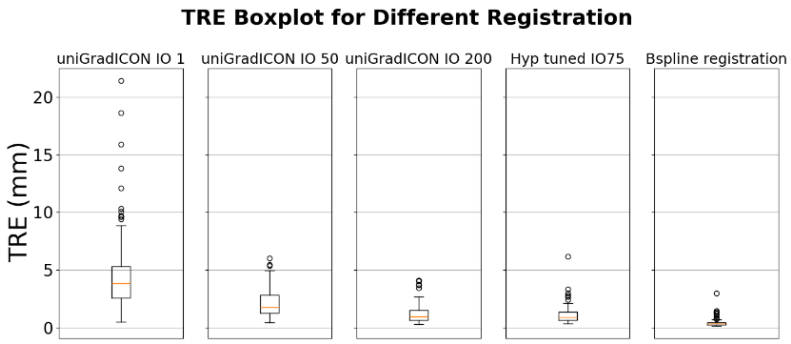


Figure 3: Dice score boxplots comparing TRE of uniGradICON model at different IO and B-spline deformable registration.

Our hyperparameter search identified an optimal configuration consisting of a learning rate of 0.0004, LNCC Gaussian kernel sigma of 2, and 75 IO iterations. This optimized configuration achieved a mean TRE of 1.11 mm (SD=0.76 mm, 95% CI [0.96-1.25 mm]), which performed better than the default hyperparameters with 200 iterations ($p<0.001$), but still worse than that achieved by B-spline registration ($p<0.001$).

3.2 Computational Efficiency

Computational speed significantly differed between the two-registration approaches. The B-spline method took more than 10 minutes per registration when executed on a 32 core CPU system. In contrast, uniGradICON took on average 3.5 minutes for 200 IO and 1.3 minutes with hyperparameter tuned settings using 75 IO while running on a Nvidia A40 GPU. These results show a 7-10X speedup for uniGradICON in comparison to B-spline registration. The faster execution time of uniGradICON can be attributed to the combination of efficient neural network inference running on a GPU and the elimination of iterative optimization of transformation parameters, which is replaced by small number of gradient steps for instance optimization. It is worth noting that zero-shot uniGradICON inference (without any instance optimization) runs even faster, though our results demonstrate that some degree of instance optimization is necessary to achieve accurate registrations.

Figure 4 illustrates examples of multi-step B-spline registration alongside several cases of foundational AI-based registration. For each pair, the post procedure slice containing the ablation zone and its corresponding pre-procedure slice are shown prior to and post image registration. In the overlay images, the registered pre-treatment scan is displayed in red and the post-treatment scan in blue and overlap between the two results in grey/white which show regions of aligned overlap. Figures 4a and 4b show B-spline and foundational AI-based registration for the same case, where the registered image shows accurate alignment between the anatomy including blood vessels and chest wall. The registered tumor (red) is within the ablation zone (blue). The B-spline registration is observed to have better vessel overlap (white) in the region surrounding the ablation zone over uniGradICON. Figures 4c, 4d, and 4e present additional ablation cases where the anatomy in the pre and post ablation scans align well, demonstrating successful registration. Figure 4f involves a substantially larger ablation region than the other cases, where the registration was more challenging. Overall spatial correspondence between pre- and post-treatment scans was still successfully achieved, with marginally reduced registration accuracy observed in regions adjacent to the tumor and surrounding vasculature, likely reflecting the inherent complexity in such challenging cases.

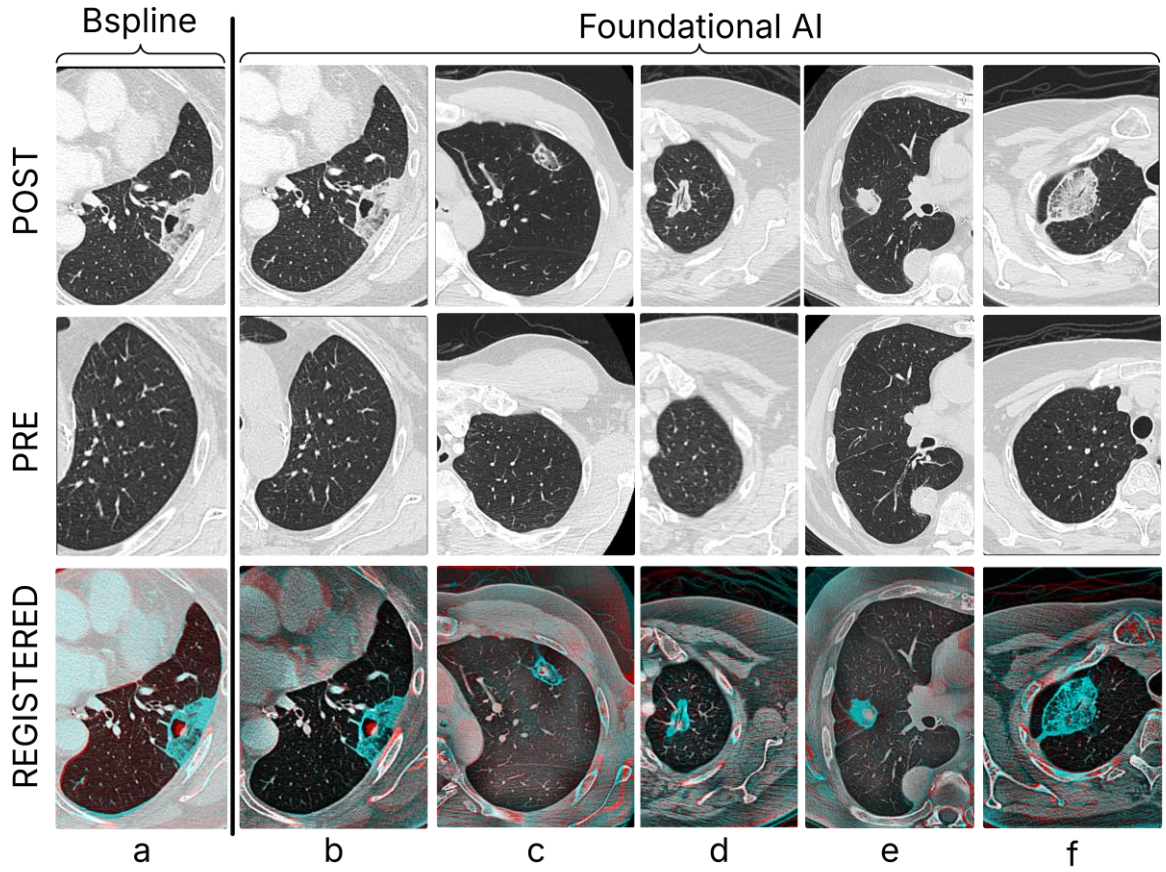


Figure 4: Post-treatment scan slice with ablation zone (row 1), corresponding unregistered pre-treatment scan slice (row 2), and registered overlay of pre- (red) and post- treatment (blue) slices (row 3); (a) and (b) compare B-spline and uniGradICON, while (c)-(d) show several examples of uniGradICON registrations.

4. DISCUSSION

This study performed a direct comparison of B-spline registration and foundational AI-based uniGradICON framework for lung microwave ablation CT registration. The results reveal an accuracy-speed tradeoff: B-spline registration achieved higher accuracy (TRE 0.43 mm) at substantially higher computational cost (7-10X higher), while the optimized uniGradICON executed significantly faster but with a 2.5-fold reduction in accuracy (1.11 mm). Although both errors are small in absolute terms, this difference may be clinically relevant depending on the application.

4.1 Impact of Instance Optimization

The results highlight the importance of instance optimization for achieving accurate registrations with uniGradICON. Zero-shot inference resulted in a mean TRE of 4.66 mm, which limits its application to lung ablation margin analysis. Applying instance optimization with 75–200 iterations and hyperparameter tuning reduced the error to approximately 1.1–1.2 mm, corresponding to an improvement of roughly 75%. Further improvement may potentially be achieved through data specific fine-tuning.

4.2 Factors Contributing to Performance Difference

Several factors may have contributed to the observed performance gap between B-spline and uniGradICON. First, uniGradICON was applied without task-specific fine-tuning and relies solely on pre-training across diverse registration tasks. Fine-tuning on lung ablation CT data would likely improve performance but would require additional computational effort and access to a sufficiently large, well-annotated lung ablation dataset.

Second, differences in effective spatial resolution may play a role. Due to GPU memory limitations, uniGradICON internally operates at reduced resolution of 175x175x175 voxels, whereas classical B-spline methods can process full-resolution images on the CPU. This difference may potentially affect the alignment of fine anatomical structures, such as small vessels, which are commonly used as landmarks for TRE evaluation.

Finally, lung ablation registration presents challenges that may inherently favor classical optimization-based approaches. Thermal ablation induces localized tissue damage, edema, and structural distortion, leading to complex, spatially confined deformation patterns. Such case-specific deformations may be difficult for a pre-trained neural network to generalize, whereas iterative optimization can adapt directly to the deformation characteristics of each individual case.

5. CONCLUSION

Our B-spline registration demonstrated superior accuracy compared to uniGradICON, though at 7 to 10 times higher computational cost. These results reveal an accuracy-speed tradeoff between the two approaches. Notably, while hyperparameter tuning of uniGradICON on task-specific data meaningfully improved performance, accuracy remained lower than that of B-spline registration. These findings motivate a larger-scale study with more data, careful control of variables, and task-specific model fine-tuning of uniGradICON for a rigorous comparison between the two approaches.

REFERENCES

1. Moussa AM, Ziv E, Solomon SB, Camacho JC. Microwave ablation in primary lung malignancies. Thieme Medical Publishers; 2019:326-333.
2. Mohn F, Pal K, O'Connor C, et al. Association between three-dimensional margin assessment and local tumor control in percutaneous lung cryoablation of colorectal metastases. *Radiology: Imaging Cancer*. 2026;8(1):e250113.
3. Lin Y-M, Paolucci I, Albuquerque Marques Silva J, et al. Ablative margin quantification using deformable versus rigid image registration in colorectal liver metastasis thermal ablation: a retrospective single-center study. *European radiology*. 2024;34(9):5541-5550.
4. Rueckert D, Sonoda LI, Hayes C, Hill DL, Leach MO, Hawkes DJ. Nonrigid registration using free-form deformations: application to breast MR images. *IEEE transactions on medical imaging*. 2002;18(8):712-721.
5. Zou J, Gao B, Song Y, Qin J. A review of deep learning-based deformable medical image registration. *Frontiers in Oncology*. 2022;12:1047215.
6. Tian L, Greer H, Kwitt R, et al. unigradicon: A foundation model for medical image registration. Springer; 2024:749-760.
7. Geevarghese R, Alexander ES, Elsakka A, et al. Outcomes following microwave ablation of 669 primary and metastatic lung malignancies. *European Radiology*. 2025:1-10.
8. Keshavamurthy KN, Eickhoff C, Ziv E. Pre-operative lung ablation prediction using deep learning. *European Radiology*. 2024;34(11):7161-7172.
9. Brett M, Leff AP, Rorden C, Ashburner J. Spatial normalization of brain images with focal lesions using cost function masking. *Neuroimage*. 2001;14(2):486-500.
10. Pieper S, Halle M, Kikinis R. 3D Slicer. IEEE; 2004:632-635.
11. Tian L, Greer H, Vialard F-X, et al. Gradicon: Approximate diffeomorphisms via gradient inverse consistency. 2023:18084-18094.
12. Ntatsis K, Dekker N, Van Der Valk V, et al. itk-elastix: Medical image registration in Python. *Scipy 2023*. 2023;
13. Team G, Anil R, Borgeaud S, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:231211805*. 2023;